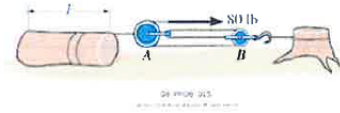


## General Friction Problems from Old Exams

### Practice Exam A:

#### Section 3 – Friction

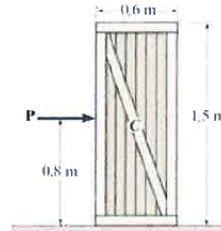
3.1 (5 pt) The log has a coefficient of static friction of  $\mu = 0.3$  with the ground and a weight of 40 lb/ft. If a man can pull on the rope with a maximum force of 80 lb, determine the greatest length "L" of the log he can bring to the point of impending motion.



$T = 320$   
 $.3N = 320$   
 $N = 1066.7 \text{ lb} = W = L \cdot 40$   
 $L = 26.7 \text{ ft}$

3.2 (4 pt) What value of P would cause tipping of the uniform 60-kg crate?

$60(9.81) = 588.6 \text{ N}$   
 $+ \sum M_N = 0;$   
 $P(.8) - 588.6(.3) = 0$   
 $P = 220.7 \text{ N}$

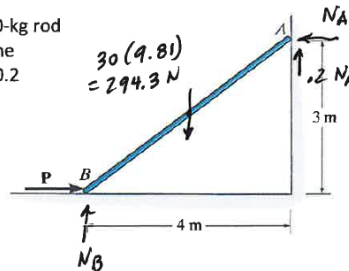


### Practice Exam B:

#### Section 3 - Friction

12. (5 pt) Determine the minimum force P to prevent the uniform 30-kg rod AB from sliding down. The contact surface at B is smooth, whereas the coefficient of static friction between the rod and the wall at A is  $\mu_s = 0.2$

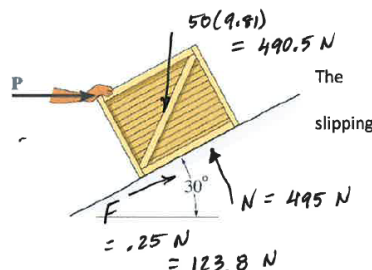
- A. 143 N
- B. 155 N
- C. 138 N
- D. 129 N
- E. 117 N



$+ \sum M_B = 0; -294.3(2) + N_A(3) + .2N_A(4) = 0$   
 $N_A = 154.9 \text{ N}$   
 $P = N_A$

13. (5 pts) Determine the minimum horizontal force P required to prevent the crate from slipping down the plane. The crate has a mass of 50 kg and the coefficient of static friction between the crate and the plane is  $\mu_s = 0.25$ . At the point of impending motion, the normal force at the contact surface is 495 N.

- A. 140 N
- B. 203 N
- C. 449 N
- D. 474 N
- E. 335 N



$P \cos 30 - 490.5 \sin 30 + 123.8 = 0$   
 $P = 140.2 \text{ N}$

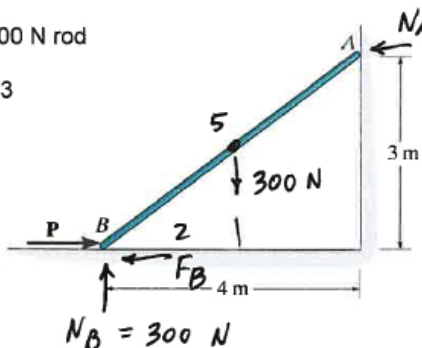
## Practice Exam Spring 2022

### Section 3 - Friction

14. Determine the minimum force  $P$  to initiate motion of the uniform 300 N rod  $AB$  up the wall. The contact surface at  $A$  is smooth, whereas the coefficient of static friction between the rod and the floor at  $B$  is  $\mu_s = 0.3$

- A. 90 N
- B. 190 N
- C. 290 N ✓
- D. 300 N
- E. 390 N

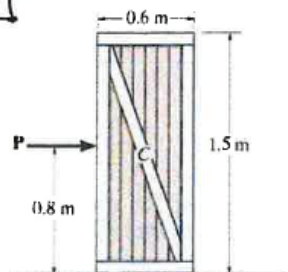
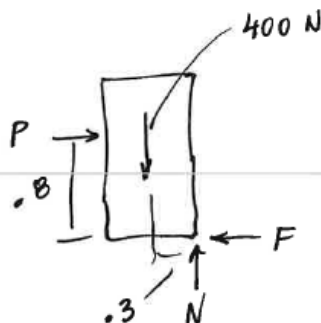
By inspection:  $N_B = 300 \text{ N}$   
 Then,  $F_B = .3(300) = 90 \text{ N}$   
 $+\circlearrowleft \sum M_B = 0; 300(2) - N_A(3) = 0$   
 $N_A = \frac{600}{3} = 200 \text{ N}$



Finally,  $+\rightarrow \sum F_x = 0; P - 90 - 200 = 0; \boxed{P = 290 \text{ N}}$

15. Determine the minimum horizontal force  $P$  required to cause the crate to move. The crate weighs 400 N and the coefficient of static friction between the crate and the plane is  $\mu_s = 0.4$ .

- A. 100 N
- B. 120 N
- C. 130 N
- D. 150 N ✓
- E. 180 N



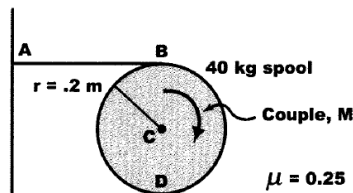
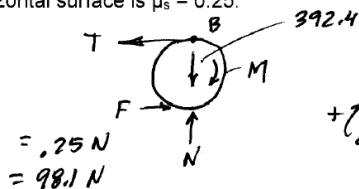
Slip:  $F = .4(400) = \boxed{160 \text{ N} = P_{\text{slip}}}$

Tip:  $+\circlearrowleft \sum M_N = 0; P(.8) - 400(.3) = 0;$   
 $\boxed{P = 150 \text{ N}} \sim \text{Will tip first,}$

### Spool Problem with Couple

2. A cable is wrapped around a spool and is attached to the wall at A. A couple  $M$  is applied to the spool as shown. Determine the minimum couple  $M$  required to cause the 40 kg spool to move. The coefficient of static friction between the spool and the horizontal surface is  $\mu_s = 0.25$ .

- A. 39.2 N-m
- B. 10.4 N-m
- C. 19.3 N-m
- D. 29.5 N-m
- E. 49.1 N-m



$+\circlearrowleft \sum M_D = 0; M - 98.1(.4) = 0$   
 $\boxed{M = 39.2 \text{ N}\cdot\text{m}}$

### Spool Problem with Force on Cable

15. (5 pt) A cable is wrapped around a 39 lb spool as shown. An upward force  $P$  is applied to the cable. The wall at B is smooth, and the coefficient of static friction between the spool and the horizontal surface is  $\mu_s = 0.3$ . When the spool is on the verge of slipping at A, the normal force  $N_B$  at B is 9 lb. Determine the force  $P$ .

- A. 23 lb
- B. 39 lb
- C. 13 lb
- D. 20 lb
- ✓ E. 9 lb

